

An Unconditional 67.3399% Bound and Conditional Advances Beyond 67.92% for Simple Critical Zeros of the Riemann Zeta Function

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23 August 2026, 00:50:38 PDT
Version 3

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DOI: 10.5281/zenodo.22066689

Abstract

Let $N(T, 2T)$ count the nontrivial zeros of the Riemann zeta function in $T < \text{Im } \rho \leq 2T$, with multiplicity, and let $N_{0,s}(T, 2T)$ count those zeros that are both simple and on the critical line. We prove the unconditional numerical bound

$$\liminf_{T \rightarrow \infty} \frac{N_{0,s}(T, 2T)}{N(T, 2T)} \geq \frac{673399}{10^6} = 0.673399. \quad (1)$$

Because every zero counted by $N_{0,s}$ is simple, distinct, and on the critical line, the same unconditional 0.673399 bound holds for all three corresponding proportions.

This strictly improves the optimized 0.672500703679... bound obtained from the Montgomery–Taylor finite-compression method. The proof combines four admissible bandlimited profiles and five rows of nonnegative pair interactions with a rational piecewise-linear subaction, directed Arb enclosures on all 243 compact transition domains, and exact reset, endpoint, and tail ledgers. We also prove that the fixed optimal Montgomery–Taylor kernel with pure Gram/rank machinery cannot force a bound above 0.6736, even through arbitrarily large fixed blocks or their thermodynamic limit.

Separately, we strengthen the author’s earlier conditional 0.6792 theorem. The uniform vanishing-box hypothesis may be replaced by zero-density convergence or, at fixed bounded depth, by density convergence to a common symmetric horizontal profile, possibly nonzero. Retaining the discarded nonnegative off-diagonal energy then gives $0.6792 + \varepsilon_f$ for an effective witness-dependent $\varepsilon_f > 0$. These results remain conditional and do not prove the Riemann hypothesis.

Certificate note. The exact range-5 computation closed at 2026-08-23T00:50:38-07:00 (2026-08-23T07:50:38Z). Its 6,052-byte path-neutral certificate core records PASS and binds every required finite inventory. The compact core is included in the accompanying archival package. The full formal replayable numerical package is in preparation.

1. Introduction

Montgomery's pair-correlation method [10], under the Riemann hypothesis (RH), gives a lower bound of two thirds for the proportion of simple zeros of the Riemann zeta function. Montgomery and Taylor improved that number, and Cheer and Goldston [4] recorded the optimized value

$$P = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot\left(\frac{1}{\sqrt{2}}\right) = 0.6725007036794116457\dots \quad (2)$$

Chirre, Goncalves, and de Laat (CGdL) [3] later used semidefinite programming to obtain 0.6792, still under RH. Baluyot, Goldston, Suriajaya, and Turnage-Butterbaugh (BGSTB) [2] proved an unconditional complex-difference form of Montgomery's theorem. Claude [5] then replaced the RH-dependent zero-side reading by finite-compression inertia and obtained the constant P unconditionally for simple critical-line zeros.

The first purpose of this paper is to show that P is not the endpoint of that unconditional framework. Its abstract rank-trace inequality treats the sampling vectors as arbitrary. In fact they are translates of bandlimited profiles, and their correlations retain geometric information. We average four fixed profile inequalities, preserve five rows of nonnegative pair interactions, and solve the resulting continuous stationary gap problem by a finite-memory subtraction and a renewal decomposition. The completed chain gives (1). We also retain the periodic Gram witness proving the 0.6736 ceiling for the earlier fixed-kernel pure-Gram mechanism.

The second purpose is to strengthen the author's earlier paper [6], which combined the BGSTB complex-difference theorem with the CGdL certificate. That paper proved

$$\max_{T < \gamma \leq 2T} |\beta - \tfrac{1}{2}| \log T \longrightarrow 0 \implies \liminf \frac{N_{0,s}(T, 2T)}{N(T, 2T)} \geq 0.6792. \quad (3)$$

Its central device was an integrated-kernel transfer: endpoint vanishing and two integrations by parts compared the complex-difference quadratic form with its ordinate projection without asserting pointwise convergence [6, Sections 3–5]. We use exactly that kernel and projection, but retain the signed Hilbert-space covariance that the absolute-value proof in [6] discarded. The new hypothesis is

$$\#\{\rho : T < \gamma \leq 2T, |\beta - \tfrac{1}{2}| \log T \geq \varepsilon\} = o(N(T, 2T)) \quad (\varepsilon > 0 \text{ fixed}). \quad (\text{BOX-DENS})$$

We introduce this microscopic density condition under the name BOX-DENS. A second condition, PROFILE-DENS(B), allows the horizontal zero profile at each ordinate to converge in density to any common symmetric probability measure on $[-B, B]$, rather than to the point mass at zero. We prove that PROFILE-DENS(B) is strictly weaker than bounded BOX-DENS and, for an effective witness-dependent constant $\varepsilon_f > 0$,

$$\boxed{(\text{BOX-DENS or PROFILE-DENS}(B)) \implies 0.6792 + \varepsilon_f.} \quad (4)$$

This is a genuine weakening of (3). It allows exceptional zeros of fixed or growing normalized depth, provided they have density zero. It also allows a positive proportion of off-line zeros if their normalized depths tend to zero. PROFILE-DENS(B) goes farther: every ordinate fiber may carry the same fixed nonzero normalized-depth profile. Neither condition is known unconditionally. The

value of ε_f is effective in principle; the theorem below proves that it is strictly positive, while a useful directed decimal is left to a separate finite computation.

The two results should be kept logically separate. Equation (1) is an unconditional numerical improvement. Equation (4) both weakens the hypothesis and strictly improves the conclusion of [6], but remains conditional. The two results extend the earlier paper in two independent directions without revising or replacing its historical claim.

Current certification and frontier snapshot

The status labels in this table are part of the mathematical statement. A larger decimal is not automatically a stronger theorem.

Value	Meaning	Status on 23 August 2026
(0.6725007036794116...)	optimized Montgomery–Taylor constant	external published benchmark
(0.6729130507261694...)	previously released project lower bound	proved and released
(0.673399)	headline proved in this paper	certified range-5 theorem; compact core PASS
(0.6738805731772266)	top range-15 survey candidate	numerical contender only
(0.673996)	Josephs Number, the cap on the present scalar stationary-ensemble process	exact method ceiling proved in this project

For the 0.673399 computation, exact directed certificates cover all 243 continuous compact transitions: 188 close directly and 55 close through hierarchical exact subbox replacements. The smallest certified transition lower bound is positive. A separate mesh-directed calculation closes all 160 short finite-reset roots. The canonical 81-state seed set, its exact Dijkstra credits, 324 long-reset roots, 121 endpoint bounds, large-gap tail, and final root have also passed and are bound by the compact certificate core identified on the first page.

The Josephs Number is a ceiling on a specified method, not a claim about the true proportion of zeta zeros. Likewise, the range-15 value is a search result, not a theorem. These distinctions are explicit here.

2. The finite-compression inequality

We follow the normalization of Claude [5]. Let $P_1 \succeq 0$ be the simple-critical-line contribution to the normalized finite Weil compression, so $P_1 = VV^*$, with the columns of V equal to the simple-zero sampling vectors. Write $\hat{A} = P_1 + Q$, where $Q = Q_+ - Q_-$, $Q_{\pm} \succeq 0$, $Q_+Q_- = 0$, and $\text{rank } Q_+ \leq b$; here b is the positive-remainder rank budget in [5, Proposition 4.4]. Pad the eigenvalues of P_1 with zeros to $p_1, \dots, p_s$, where s is the number of simple line zeros in the window. Define

$$f(x) = \begin{cases} (x-1)^2, & 0 \leq x \leq 2, \\ 2x-3, & x \geq 2. \end{cases} \quad (5)$$

The rank–trace argument admits the exact strengthening

$$\|P_1 + Q\|_F^2 \geq 2 \operatorname{tr} P_1 - s + 4 \operatorname{tr} Q - 4b + \sum_{j=1}^s f(p_j). \quad (6)$$

Indeed, von Neumann's trace inequality reduces the adverse interaction to scalar pairs $p, n \geq 0$, and

$$\min_{n \geq 0} \{(p - n)^2 + 4n\} = \begin{cases} p^2, & p \leq 2, \\ 4p - 4, & p \geq 2, \end{cases} = 2p - 1 + f(p). \quad (7)$$

Every positive eigenvalue q of Q_+ satisfies $q^2 \geq 4q - 4$, proving (6). With the same counting substitution as in [5, Proposition 4.4], and with $K = V^*V$ the Gram matrix of the simple-zero sampling vectors, this becomes

$$s \geq 4 \operatorname{tr} \hat{A} - 2N - \|\hat{A}\|_F^2 + \operatorname{tr} f(K). \quad (8)$$

The nonzero spectra of VV^* and V^*V agree, and $f(0) = 1$, so the padded spectral sum in (6) is exactly $\operatorname{tr} f(K)$. The trace asymptotics in [5] turn the first three terms of (8) into $PN + o(N)$:

$$s \geq PN + \operatorname{tr} f(K) + o(N). \quad (9)$$

The earlier proof discarded the nonnegative last term. The present unconditional argument extracts a uniform linear contribution from it.

3. The four-profile range-5 energy

We apply (9) separately to four fixed admissible profiles $p_1, \dots, p_4$, multiply the resulting inequalities by nonnegative rational weights θ_j with $\sum_j \theta_j = 1$, and only then add them. Thus

$$s \geq \bar{P}N + \sum_{j=1}^4 \theta_j \operatorname{tr} f(K_{p_j}) + o(N), \quad \bar{P} = \sum_{j=1}^4 \theta_j P(p_j). \quad (10)$$

The nonlinear penalties in (10) are evaluated before averaging; there is no Gram inequality for an averaged profile and no duplicated rank charge. The profiles are the even cosine polynomials

$$p_j(t) = 1 + \sum_{n=1}^{80} a_{j,n} \cos(2\pi n t), \quad -\frac{1}{2} \leq t \leq \frac{1}{2}, \quad (11)$$

with coefficients and weights rationalized at denominator 10^{12} . The four weight numerators are

$$36287820623, \quad 421033835435, \quad 202969544591, \quad 339708799351. \quad (12)$$

They sum to 10^{12} . Directed Lipschitz estimates certify positivity of every profile, with the smallest recorded lower margin greater than 0.8264, and directed evaluation gives

$$\bar{P} = 0.6721191688908208431825544112 \dots \quad (13)$$

Write

$$k_j(x) = \int_{-1/2}^{1/2} p_j(t) e^{ixt} dt, \quad q(x) = 2c \sum_{j=1}^4 \theta_j |k_j(x)|^2 \geq 0, \quad (14)$$

and fix the exact parameters

$$c = \frac{6740037}{10^7}, \quad \eta = \frac{2559511110}{10^{12}}, \quad C_* = \frac{673399}{10^6}. \quad (15)$$

For a one-sided sequence of nonnegative normalized gaps $g = (g_i)_{i \geq 0}$, retain five forward interaction rows:

$$F_5(g) = c + \frac{\eta}{2\pi} g_0 + \sum_{r=1}^5 q(g_0 + \dots + g_{r-1}). \quad (16)$$

The stationary floor required for the headline bound is

$$\beta_* = 2c - \frac{c\bar{P} - \eta}{C_*} = 0.6790855638176799425865 \dots, \quad C_* = \frac{c\bar{P} - \eta}{2c - \beta_*}. \quad (17)$$

The rational alphabet used to construct the finite-memory subaction is

$$S = \frac{6496123}{10^6}, \quad L = \frac{6138729}{500000}, \quad X = \frac{3637}{200}. \quad (18)$$

It has $3^4 = 81$ memory states and $3^5 = 243$ transitions. The exact state potential combines a denominator- 10^{15} tabular potential, rational clipped affine jets, and rational piecewise-linear functions of contiguous state sums. Its transport manifest records the debit from absorbing an earlier rational offset into the tabular potential. The complete profile coefficients, weights, potential tables, and normalization ledger accompany the certificate.

4. Compactification and the directed Arb certificate

The local energy admits an exact symmetrization by a finite-memory coboundary. If an interaction of length r is averaged over its $6 - r$ positions in a five-gap word, then

$$F_5^{\text{sym}}(g) = c + \frac{\eta}{10\pi} \sum_{a=0}^4 g_a + \sum_{r=1}^5 \frac{1}{6-r} \sum_{a=0}^{5-r} q(g_a + \dots + g_{a+r-1}). \quad (19)$$

There is an explicit finite-memory function J for which $F_5^{\text{sym}} - F_5 = J \circ \sigma - J$. Let H denote the total exact state correction: its tabular, clipped-affine, and piecewise-linear parts are all included. The compact transition obligation is

$$F_5^{\text{sym}} + H \circ \sigma - H - \beta_* \geq 0. \quad (20)$$

4.1 Closed compact transition gate

Each of the 243 symbolic transitions represents a product of rational compact gap intervals. The verifier evaluates the smooth pair field with directed Arb arithmetic and the state correction with exact rational arithmetic. Away from piecewise-linear crease intersections it uses value and derivative bounds. At hard intersections it aggregates terms by contiguous sum, represents the resulting half-space geometry with exact difference constraints, and proves an affine minorant on every retained cell. Ordinary rational subdivision serves as the final fail-closed fallback.

All 243 transition domains have passed. Of these, 188 close directly and 55 close by exact hierarchical replacement. The smallest full-cover lower bound is

$$\frac{35462857173642647474964770625129719976195638840695703}{822752278660603021077484591278675252491367932816789931674304512} > 4.31027 \times 10^{-11}.$$

The unified continuous-cover receipt, identified in the accompanying certificate core, binds the 81-state, 243-transition universe and the exact potential and source identities. This is a completed local certificate, but by itself it does not prove (1).

4.2 Exact finite-word closure

Put $b = \eta/(2\pi)$. For a four-gap state $g = (g_0, \dots, g_3)$, define

$$A_L(g) = 5(c - \beta_*) + \sum_{i=0}^3 \frac{4-i}{5} b g_i + \sum_{r=1}^4 \sum_{i=0}^{4-r} \frac{5-r-i}{6-r} q(g_i + \dots + g_{i+r-1}),$$

and

$$J_L(g) = \sum_{i=0}^3 \frac{i+1}{5} b g_i + \sum_{r=1}^4 \sum_{i=0}^{4-r} \frac{i+1}{6-r} q(g_i + \dots + g_{i+r-1}).$$

The seed for state s is the directed infimum of $A_L + H_s$ over its component box. Exact integer Dijkstra propagation then relaxes

$$p_{\text{dst}} \leftarrow \min\{p_{\text{dst}}, p_{\text{src}} + \ell_e\},$$

where ℓ_e is the downward-rounded certified loss of compact transition e . For a finite outgoing reset a , set

$$\chi_4(a, g) = ba + \sum_{j=0}^4 q(a + g_j + \dots + g_3),$$

where the empty suffix at $j = 4$ contributes $q(a)$. The 324 long-reset obligations are

$$p_s + \chi_4(a, g) + J_L(g) - H_s(g) \geq 0$$

on 81 state boxes and four reset branches.

Short components contain $0 \leq \ell \leq 3$ compact gaps. Their exact reset inequality consists of $(\ell + 1)(c - \beta_*)$, the span charge on all compact gaps and the reset, every internal pair term, and all reset-plus-suffix pair terms. All 160 such domains have passed a directed derivative-mesh verifier; the smallest certified lower is greater than 2.29×10^{-6} .

For terminal components, let B_{short} be the largest of the 40 short endpoint deficits and let

$$B_{\text{long}} = -\min_s \inf_g \{p_s + J_L(g) - H_s(g)\}, \quad B = \max\{0, B_{\text{short}}, B_{\text{long}}\}.$$

At a reset $a \geq 30$, nonnegative pair terms crossing the cut may be dropped, while the span ledger contributes at least $30b = 0.01222076535165 \dots$. The endpoint calculation gives

$$B \leq \frac{9863684272393059149187}{10^{24}} = 0.009863684272393059149187,$$

so the directed tail margin is greater than 0.00235708107926065665 . Thus $30b > B$ pays every nonterminal component, leaving only one terminal defect B . Division by word length and finite-word exhaustion then give the stationary floor β_* .

The canonical 81 seed receipts, propagated exact-integer Dijkstra credits, 324 long-reset receipts, 40 short and 81 long endpoint receipts, tail inequality, and final root have all passed. The compact certificate core binds their roles, sizes, and SHA-256 identities together with the frozen candidate, ensemble, potential, and verifier-source identities. The compact core is included in the accompanying archival package and freezes the completed computation and its identity chain; the full formal replayable numerical package is in preparation.

5. From the stationary floor to the spectral penalty

The stationary certificate concerns the mixed quadratic pair energy, whereas (10) contains the clipped convex spectral penalties. Their relation must be used in the correct direction: for a correlation matrix G , $\text{tr } f(G) \leq \|G - I\|_F^2$ in general, with equality only while the spectrum remains in the quadratic branch. The transfer is supplied by the following proposition.

Proposition 1 (stationary spectral transfer). Let a finite ensemble of admissible profiles have mixed baseline $\bar{P}$, target parameter c , span price η , and nonnegative pair potential q as in (10)–(16). Suppose a finite-memory directed certificate and renewal ledger prove the stationary floor β . Then the corresponding finite compressions satisfy

$$\liminf_{T \rightarrow \infty} \frac{s}{N} \geq \frac{c\bar{P} - \eta}{2c - \beta}.$$

Proof. Apply the rank–trace inequality before taking a stationary limit. On every quadratic component, the mixed Frobenius pair ledger is exactly the sum of the four spectral penalties. If a component exits the quadratic branch, the fixed-block branch guard

$$\mathrm{tr} f(G) \geq \min\{\|G - I\|_F^2, m/(m-1)\}$$

and the smallest ensemble weight pay the corresponding local deficit. The coboundary terms telescope. At a renewal cut, nonnegative cross-cut pairs may be discarded and the full outgoing span charge is assigned exactly once; the reset inequality pays every nonterminal component. What remains is one uniform endpoint defect. Summing the component inequalities, dividing by the number of zeros, and then taking the finite-volume limit removes that defect and gives

$$(2c - \beta)s \geq (c\bar{P} - \eta)N + o(N).$$

This is the asserted inequality. $\square$

The application of Proposition 1 uses the exact data in (13)–(20). Every finite inventory listed in Section 4 has passed and is bound by the compact certificate core.

Theorem 1. Unconditionally,

$$\liminf_{T \rightarrow \infty} \frac{N_{0,s}(T, 2T)}{N(T, 2T)} \geq \frac{c\bar{P} - \eta}{2c - \beta_*} = \frac{673399}{10^6} = 0.673399.$$

No RH, vertical-box, zero-density, or pair-correlation conjecture enters the proof. The word “unconditional” has the same scope as the trace asymptotics of [5] on which (9) rests.

Corollary 1. Let $N_s(T, 2T)$ count all simple zeros, $N_{\mathrm{dist}}(T, 2T)$ count distinct zeros, and $N_0(T, 2T)$ count critical-line zeros with multiplicity. Then

$$\min \left\{ \liminf_{T \rightarrow \infty} \frac{N_s(T, 2T)}{N(T, 2T)}, \liminf_{T \rightarrow \infty} \frac{N_{\mathrm{dist}}(T, 2T)}{N(T, 2T)}, \liminf_{T \rightarrow \infty} \frac{N_0(T, 2T)}{N(T, 2T)} \right\} \geq 0.673399.$$

This is immediate from $N_{0,s} \leq N_s, N_{\mathrm{dist}}, N_0$.

A certified ceiling for the fixed-kernel pure-Gram route

The larger-block construction has a sharp limitation that is useful to state alongside the improvement. Let R be the compactly supported Fourier density of k^2 . Writing $C = 2 \sin(\sqrt{2}/2)/\sqrt{2}$, it is even, supported on $[-1, 1]$, and for $0 \leq u \leq 1$ is

$$R(u) = \frac{\sin(\sqrt{2}(1-u))/\sqrt{2} + (1-u) \cos(\sqrt{2}u)}{2C^2}.$$

Consider a periodic q -point cell with normalized density $c = 2\pi q/H$, offsets $0 \leq \tau_j < 1$, and $S_\ell = \sum_j e^{2\pi i \ell \tau_j}$. Poisson summation gives its exact per-particle Frobenius energy

$$E(c, \boldsymbol{\tau}) = cR(0) - 1 + \frac{2c}{q^2} \sum_{1 \leq \ell \leq \lfloor q/c \rfloor} R(c\ell/q) |S_\ell|^2. \quad (20a)$$

For a certified rational 33-point cell at $c = 421/625 = 0.6736$, a 256-bit Arb evaluation of the finite sum proves

$$E = 0.00162820394118948401298\dots, \quad c(1 - E) - P > 2.5381458031 \times 10^{-6}. \quad (20b)$$

Since $f(\lambda) \leq (\lambda - 1)^2$ for every $\lambda \geq 0$, every finite Gram matrix satisfies $\text{tr } f(G) \leq \|G - I\|_F^2$. Long finite segments of the periodic cell have $s/N \rightarrow c$ and $s^{-1}\|G - I\|_F^2 \rightarrow E$. They are therefore consistent with the master inequality (9) at density c . This proves the following method statement.

Proposition 2. The fixed optimal Montgomery–Taylor kernel together with the rank inequality (9) and any pure simple-Gram lower bound cannot force a proportion greater than 0.6736. This includes arbitrary fixed block sizes, their thermodynamic limit, and valid mixtures of block partitions.

This is a ceiling on the method, not an upper bound on the true proportion of simple critical zeros. The rational cell and directed interval output are included in the accompanying archival source package.

Consequently, within the fixed-kernel component/chunk certificate architecture of version 1.1, improvements toward 0.6736 are computational refinements of the same analytic mechanism. Proposition 2 shows that crossing 0.6736 cannot come from such a refinement alone: it requires new analytic input or an enlargement of the admissible technique space. The profile ensemble in Sections 3–5 is such an enlargement. This statement is specific to the fixed-kernel architecture and makes no claim about other approaches to simple critical zeros.

6. The earlier integrated-kernel transfer

We now return to the second result. This section deliberately recalls the machinery of the author’s earlier paper [6], both to make the strengthening standalone and to identify exactly where the old uniform hypothesis is replaced.

Let f_C be the fixed interval-certified CGdL witness used in [6], let R be its last sign-change radius, and put

$$F(\alpha) = f_C(R\alpha), \quad K(z) = \int_{-1}^1 F(\alpha) e^{\alpha z} d\alpha, \quad W(z) = \frac{4}{4 - z^2}. \quad (21)$$

The endpoint identity $F(\pm 1) = 0$ is the source of the two integrations by parts in [6, Section 4]. It yields the global estimate

$$|K(h + iy) - K(iy)| \ll_f \frac{|h|e^{|h|}}{1 + y^2}. \quad (22)$$

The earlier paper applied (22) to both endpoints of every ordered zero pair, leading to a pointwise depth envelope. Here we instead use its exact positive Hilbert-space representation. Define

$$\Psi_T(z) = K(z)W(z/L) = \int_{\mathbb{R}} e^{\lambda z} d\mu_T(\lambda), \quad (23)$$

valid throughout the present range $|\text{Re } z| \leq L/2$, where μ_T is a finite positive measure. Here positivity follows from the nonnegative CGdL witness F and $W(z) = \int_{\mathbb{R}} e^{-2|t|} e^{tz} dt$: convolution and pushforward under $\lambda = \alpha + t/L$ give the positive measure in (23). At each ordinate γ , let U_γ be the symmetric multiset of horizontal coordinates $u = (\beta - 1/2)L$, let $m_\gamma = |U_\gamma|$, and put

$$D_\gamma(\lambda) = \sum_{u \in U_\gamma} e^{\lambda u} - m_\gamma = \sum_{u > 0} 2(\cosh(\lambda u) - 1) \geq 0. \quad (24)$$

In $L^2(\mu_T)$, define

$$U_T(\lambda) = \sum_{\gamma} m_\gamma e^{i\lambda\gamma L}, \quad V_T(\lambda) = \sum_{\gamma} D_\gamma(\lambda) e^{i\lambda\gamma L}. \quad (25)$$

The complex-difference sum evaluated unconditionally by BGSTB and the ordinate-projected sum bounded in [6] satisfy exactly

$$S_* = \|U_T + V_T\|^2, \quad S_{\text{in}} = \|U_T\|^2, \quad (26)$$

so

$$S_* - S_{\text{in}} = \|V_T\|^2 + 2\Re\langle U_T, V_T \rangle. \quad (27)$$

Consequently the one-sided mixed projection estimate

$$\Re\langle U_T, V_T \rangle \geq -o(N) \quad (28)$$

is sufficient to rerun [6, Sections 3–5] and obtain 0.6792. This is the only part of the previous proof that we change.

7. A one-endpoint bound and crowding moments

For each zero put

$$d_\rho = |\beta - \tfrac{1}{2}|L, \quad \ell_\rho = \sum_{\rho'} \frac{1}{1 + \{L(\gamma - \gamma')\}^2}. \quad (29)$$

The mixed term has the lower bound

$$\Re\langle U_T, V_T \rangle \geq -C_f \sum_{\rho} d_\rho e^{d_\rho} \ell_\rho. \quad (30)$$

To see this, an ordinate atom interacting with one reflected depth pair $\pm u$ contributes

$$\Psi_T(u + iy) + \Psi_T(-u + iy) - 2\Psi_T(iy).$$

Equation (22), boundedness of W for $|\operatorname{Re}(z/L)| \leq 1/2$, and direct subtraction of the two rational W -factors bound its adverse real part by $C_f u e^u / (1 + y^2)$. Summing proves (30). Unlike the absolute pairwise estimate in [6], only one endpoint carries depth, which is why the exponent is 1 rather than 2.

We also need all fixed moments of ℓ_ρ :

$$\sum_{\rho} \ell_{\rho}^q \ll_q N \quad (q \geq 1 \text{ fixed}). \quad (31)$$

Let $C_h(t)$ count zero ordinates within h of t , with multiplicity. The Riemann–von Mangoldt formula and Tsang’s fixed even moments of $S(t+h) - S(t)$, in the form quoted by Simonic, Trudgian, and Turnage-Butterbaugh [11], give

$$\int_{T-O(1)}^{2T+O(1)} C_h(t)^r dt \ll_r T(1+hL)^r \quad (L^{-1} \leq h \leq 1). \quad (32)$$

For a prescribed exponent q , choose the next available fixed even Tsang moment at least $q+1$, and use Holder’s inequality when $q+1$ is not even.

If $|t - \gamma| \leq h/2$, then $C_h(\gamma) \leq C_{3h/2}(t)$. Integrating and counting the overlap of the zero-centered intervals gives

$$h \sum_{\rho} C_h(\gamma)^q \ll_q T(1+hL)^{q+1}. \quad (33)$$

Taking $h = 2^j/L$, decomposing (29) into dyadic shells, and applying Minkowski yields (31), since $\sum_{j \geq 0} 2^{-2j} 2^j < \infty$. On wider physical shells the standard local Riemann–von Mangoldt bound supplies $O(L)$ zeros per unit interval, whereas the kernel contributes $O((Lk)^{-2})$ on the k th interval; their total is $O(1)$ pointwise.

8. BOX-DENS implies mixed projection dominance

Baluyot [1, Theorem 1.2.1] proves, for every fixed $1/2 < \theta < 4/7$,

$$N(\sigma, T) \ll_{\theta} T^{1+\theta(1-2\sigma)} \log T \quad (\sigma \geq \tfrac{1}{2}). \quad (34)$$

Functional-equation reflection converts (34), for $0 \leq u \leq L/2$, into

$$\#\{\rho : d_{\rho} \geq u\} \ll_{\theta} N e^{-2\theta u}. \quad (35)$$

For larger u the set is empty. Choose r with $1 < r < 2\theta$, and let $q = r/(r-1)$. For $M \geq 1$, unit-depth summation in (35), followed by Holder’s inequality and (31), gives

$$\sum_{d_{\rho} \geq M} d_{\rho} e^{d_{\rho}} \ell_{\rho} \ll N M e^{-(2\theta-r)M/r}. \quad (36)$$

Thus the entire deep tail is unconditionally negligible as $M \rightarrow \infty$. For $d_{\rho} < \varepsilon$, (31) with $q = 1$ gives

$$\sum_{d_{\rho} < \varepsilon} d_{\rho} e^{d_{\rho}} \ell_{\rho} \ll \varepsilon e^{\varepsilon} N. \quad (37)$$

Finally, on $\varepsilon \leq d_\rho < M$, BOX-DENS and Holder's inequality give

$$\sum_{\varepsilon \leq d_\rho < M} d_\rho e^{d_\rho} \ell_\rho \leq M e^M o(N)^{1/r} \left(\sum_\rho \ell_\rho^q \right)^{1/q} = o(N). \quad (38)$$

Let $T \rightarrow \infty$, then $\varepsilon \rightarrow 0$ and $M \rightarrow \infty$. Equations (36)–(38) make the right side of (30) $o(N)$, proving (28).

Theorem 2. Under BOX-DENS,

$$\liminf_{T \rightarrow \infty} \frac{N_{0,s}(T, 2T)}{N(T, 2T)} \geq 0.6792. \quad (39)$$

This strengthens the main theorem of [6] by replacing its uniform vanishing-box hypothesis with convergence in zero density. The shallow and deep localizations, (36)–(37), are unconditional; BOX-DENS is used only for the bounded nonzero depth band in (38).

9. Quantitative defects and common profiles

The preceding argument has a useful quantitative form. Retain the fixed CGdL last sign-change radius

$$R = 1.0038742696224231332921744410403537955 \dots$$

and define the adverse transfer defect

$$\delta_T = \frac{(S_{\text{in}} - S_*)_+}{N(T, 2T)}. \quad (40)$$

Theorem 3 (quantitative transfer). If $\limsup_{T \rightarrow \infty} \delta_T \leq \delta$, then

$$\boxed{\liminf_{T \rightarrow \infty} \frac{N_{0,s}(T, 2T)}{N(T, 2T)} \geq 0.6792 - R\delta.} \quad (41)$$

Indeed, the fixed directed CGdL certificate and the unconditional complex-difference evaluation give $RS_* \leq (1.3208 + o(1))N$, while the ordinate diagonal gives $R^{-1}N_{\text{ord}}^* \leq S_{\text{in}}$. Equation (40), followed by $N_{0,s} \geq 2N - N_{\text{ord}}^*$, proves (41). Thus a nonzero transfer error does not destroy the method; it spends an exactly priced amount of the headline.

There is also a weakening in which depth itself need not become small. For each distinct ordinate γ , let U_γ be the symmetric multiset of normalized horizontal coordinates at that ordinate, let $m_\gamma = |U_\gamma|$, and set

$$\nu_\gamma = \frac{1}{m_\gamma} \sum_{u \in U_\gamma} \delta_u.$$

For fixed $B < \infty$, define PROFILE-DENS(B) to mean that all these measures are supported on $[-B, B]$ and there are symmetric probability measures ν_T on $[-B, B]$ such that, for every fixed $\varepsilon > 0$,

$$\sum_{\gamma: W_1(\nu_\gamma, \nu_T) \geq \varepsilon} m_\gamma = o(N). \quad (42)$$

Theorem 4 (profile in density). PROFILE-DENS(B) implies $\delta_T = o(1)$, and hence the conclusion 0.6792.

To prove this, put

$$h_T(\lambda) = \int e^{\lambda u} d\nu_T(u) - 1, \quad \mathcal{R}_T = V_T - h_T U_T.$$

If $a_\gamma = W_1(\nu_\gamma, \nu_T)$, coupling the measures and applying the two-integration-by-parts kernel estimate twice in the bounded horizontal parameter gives

$$\|\mathcal{R}_T\|^2 \leq C_B \sum_\rho a_\rho \ell_\rho, \quad (43)$$

where $a_\rho = a_\gamma$ on the fiber over γ . On $a_\rho < \varepsilon$, (31) bounds the right side by $O_B(\varepsilon N)$; on the complement, (42), Holder's inequality, and the fixed crowding moments (31) give $o(N)$. Thus $\|\mathcal{R}_T\| = o(\sqrt{N})$. Since $U_T = (U_T + V_T - \mathcal{R}_T)/(1 + h_T)$ and multiplication by $(1 + h_T)^{-1}$ is a contraction, $S_{\text{in}} \leq S_* + o(N)$, as claimed.

Bounded BOX-DENS implies PROFILE-DENS(B) with $\nu_T = \delta_0$. The converse fails: every fiber may have the same fixed nonzero symmetric profile, in which case (42) holds with no exceptional fibers although BOX-DENS fails at positive density. This is the promised strict weakening. Independently, the close-pair-weighted averaged condition

$$\sum_\rho d_\rho e^{2d_\rho} \ell_\rho = o(N) \quad (44)$$

also gives zero transfer defect by the convex decoupling inequality $(A + B)e^{A+B} \leq Ae^{2A} + Be^{2B}$. It replaces control of the worst zero by a single weighted moment and permits sparse deep exceptions.

10. Retaining the discarded energy

We now improve the conditional conclusion itself. Normalize the nonnegative CGdL ordinate kernel as an even continuous function r with $r(0) = 1$. Let $\mathcal{S}_T$ be the set of ordinates carrying exactly one zero. Mirror symmetry forces that zero to be simple and on the critical line. Put $s_T = |\mathcal{S}_T|$ and

$$\mathcal{E}_T = \sum_{\substack{\gamma, \gamma' \in \mathcal{S}_T \\ \gamma \neq \gamma'}} r(L(\gamma - \gamma')) \frac{4}{4 + (\gamma - \gamma')^2}. \quad (45)$$

Every summand is nonnegative. The diagonal-only proof discarded (45). Keeping it gives, under any hypothesis with $\delta_T = o(1)$,

$$s_T \geq 0.6792N + \mathcal{E}_T - o(N). \quad (46)$$

This follows directly by restricting the nonnegative CGdL ordinate sum to the diagonal and to distinct pairs in $\mathcal{S}_T$, then applying the same counting inequality used above. It is genuine slack within the CGdL inequality, rather than a splice of two already-evaluated final bounds.

Suppose that for some integer $m \geq 2$ and constants $A, b > 0$, every nonnegative gap vector $g_1, \dots, g_{m-1}$, with $x_0 = 0$ and $x_j = g_1 + \dots + g_j$, satisfies

$$2 \sum_{0 \leq i < j \leq m-1} r(x_j - x_i) + b \sum_{j=1}^{m-1} g_j \geq A. \quad (47)$$

The standard residue-class partition into consecutive m -blocks, applied to (46), then gives

$$\liminf \frac{N_{0,s}}{N} \geq \frac{m \cdot 0.6792 - 2\pi(m-1)b}{m-A}. \quad (48)$$

The right side is strictly larger than 0.6792 exactly when

$$0.6792A > 2\pi(m-1)b. \quad (49)$$

Theorem 5 (strict conditional amplification). For the fixed published CGdL witness there is an effective constant $\varepsilon_f > 0$ such that

$$(\text{BOX-DENS or PROFILE-DENS}(B)) \implies \liminf_{T \rightarrow \infty} \frac{N_{0,s}(T, 2T)}{N(T, 2T)} \geq 0.6792 + \varepsilon_f. \quad (50)$$

The proof also explains why the gain is necessarily nonzero. The fixed kernel r is a nonzero polynomial times a Gaussian, so it has only finitely many positive real zeros; call their number z . Choose $m \geq z + 2$ and

$$G > \frac{2\pi(m-1)}{0.6792}.$$

On the compact simplex of nonnegative gap vectors with total span at most G , the pair energy in (47) has a positive minimum a . Otherwise all pair distances would be zeros of r ; relative to the first point this would require $m-1 > z$ distinct positive zeros. Choose $0 < \lambda < \min(1, m/a)$, and set

$$A = \lambda a, \quad b = \frac{\lambda a}{G}. \quad (51)$$

For span at most G , the energy alone proves (47); for span at least G , the span term does. Moreover (49) is strict by the choice of G . Thus one may take

$$\varepsilon_f = \frac{\lambda a \{0.6792 - 2\pi(m-1)/G\}}{m - \lambda a} > 0. \quad (52)$$

All quantities in (52) can in principle be enclosed from the pinned finite polynomial–Gaussian witness. No stronger decimal is claimed here; the point is the rigorous strict inequality and the reduction of its numerical evaluation to a finite directed kernel-packing certificate. With a nonzero transfer defect δ , the same argument gives the quantitative bound

$$\liminf \frac{N_{0,s}}{N} \geq \frac{m(0.6792 - R\delta) - 2\pi(m-1)b}{m-A}.$$

11. Relation to the earlier paper

The author’s previous paper [6] remains the source for the full integrated-kernel/CGdL splice: the two form factors, diagonal extraction, endpoint calculation, close-pair estimate, and conversion from N_{ord}^* to simple critical zeros are unchanged here. What changes is only the depth estimate. The proof in [6] imposed uniform smallness before taking absolute values. Theorem 2 retains the signed Hilbert decomposition (26)–(27), prices only its adverse mixed term, and thereby lowers the exponential cost from e^{2d} to e^d . Baluyot then controls the deep tail, while Tsang’s moments prevent a density-zero exceptional set from concentrating on heavily crowded ordinates. Thus [6] records the original BOX transfer, Theorem 2 proves the strictly stronger BOX-DENS transfer, Theorem 4 permits a coherent nonzero bounded-depth profile, and Theorem 5 turns the previously discarded positive energy into a strict conditional gain.

12. A complementary Kintali refinement

Let $\kappa = \liminf_{T \rightarrow \infty} N_0(T)/N(T)$, with multiplicity. Kintali [9, Theorems 1.1 and 4.12] proved $\kappa > 0.421$ and certified $c_{\text{sel}} \leq 2.122496896689520$ for $R = 13/10$. His proof of Theorem 2.2 shows that the nonzero shifted-frequency range is empty whenever the total logarithmic tuple mass $q \leq 21/29$. Hence no new off-diagonal estimate is needed to add an endpoint-vanishing degree-three component supported there. Take

$$F(q) = \left(\frac{21}{29} - q\right) \left(-\frac{1}{4} + \frac{232}{525}q\right) \mathbf{1}_{q \leq 21/29}. \quad (53)$$

Exact rational integration followed by outward Arb evaluation gives

$$2\{D_Q(A, F) - D_Q(F, D)\} + D_Q(F, F) < -4.2894618869157747307685 \times 10^{-6}. \quad (54)$$

Therefore $c_{\text{new}} < 2.122492607227633084225269231444$, and the Levinson–Conrey inequality yields

$$\kappa > 1 - \frac{\log c_{\text{new}}}{13/10} > 0.42108372604. \quad (55)$$

This exceeds the $0.4210821714609425\dots$ bound inferred from Kintali’s same certified endpoint by more than 0.0001554579 percentage points. His 8 August preprint briefly raised the publicly announced benchmark beyond the earlier 0.417293 result [13], two days before Anthropic announced its stronger finite-compression bound. We include (53)–(55) to record that advance and the reusable low- q observation; Theorem 1 supersedes it numerically. The exact low- q certificate accompanies the archival source package.

13. Scope

Theorem 1 is a strict unconditional numerical advance. Proposition 2 is a ceiling only for the fixed-kernel pure-Gram mechanism defined above; it is not an upper bound on the true proportion of simple critical zeros. Neither statement constrains horizontal zero location.

Theorems 2–5 are substantial logical advances over [6] but remain conditional. The transfer-defect theorem gives an exact loss budget; PROFILE-DENS(B) shows that coherent nonzero normalized depth is harmless; and retained off-diagonal energy makes the conditional conclusion strictly larger than 0.6792. BOX-DENS, PROFILE-DENS(B), and the averaged condition (44) are not known unconditionally, and no such theorem is claimed here. The existence proof for ε_f is effective but does not yet provide a new printed decimal.

None of these results proves RH. Novelty and priority are asserted only for the precise results and method classes stated above.

Research provenance and responsibility

Michael Devine, an independent researcher and consultant in embedded systems and edge AI with a B.A. in Mathematics from Grinnell College, carried out this work in collaboration with publicly accessible, general-purpose AI systems. The work did not rely on a private model or a purpose-built automated theorem prover.

The project required approximately 100 hours of direct human research, mathematical and algorithmic development, proof design, computation supervision, audit triage, and manuscript work.

Michael Devine formulated and directed the research program, supplied critical mathematical and computational insights, selected and revised proof strategies, adjudicated the resulting claims, coordinated independent audits, and made the publication decisions. His contributions included recognizing that certification difficulty depended on contender geometry rather than the headline alone; insisting on geometric diversity in contender searches; proposing locally closable proof objects and the Manhattan-metric interpretation of consecutive-gap sums; and directing adaptive refinement, parallel fan-out, and failure-derived regression testing. He also introduced batched alternative propagation: carrying multiple viable parameter choices through an expensive search stage and retaining those that certify, rather than repeatedly committing to and backtracking from a single choice. This strategy was introduced by Devine in the tree search; OpenAI GPT-5.6 Sol then recognized the same structure and reapplied it to management of the large parallel cloud certificate search.

The collaboration was iterative and bidirectional. Devine’s mathematical and systems insights materially changed the models’ search, proof, and implementation strategies; model-produced derivations, failed certificate attempts, numerical patterns, and audits in turn prompted new conjectures, design decisions, and refinements by Devine. The resulting argument and certificate emerged from this repeated exchange rather than from unattended text or code generation.

This collaboration also depends on the accumulated work of the mathematical community. The author acknowledges the mathematicians, living and deceased, whose definitions, theorems, examples, expositions, and computational methods form the shared intellectual inheritance from which both human researchers and general-purpose AI systems draw. Proprietary model-training corpora are not disclosed at the level needed to identify every indirect contributor; all works used directly in the argument are therefore cited individually, while this broader debt is acknowledged collectively.

AI systems contributed extensively to derivation, implementation, numerical experimentation, optimization, auditing, certificate construction and orchestration, cloud deployment, scheduling and recovery, and drafting. OpenAI GPT-5.6 Sol developed the version 1.1 consecutive-octuple certificate and assisted with the range-5 exact piecewise-linear certificate, finite-word transfer, BOX-DENS proof, and source checks. Anthropic Claude assisted with the profile-ensemble search, directed certification, workflow design, and adversarial review. Separate model agents re-derived the ledgers and audited the proofs and interval implementation. The author's earlier paper [6] has its own disclosed provenance and audit trail.

These model contributions are research provenance, not human peer review. Devine is the human author, approved the mathematical direction and claims, and assumes responsibility for the manuscript.

Author biography

Michael Devine is an independent researcher and a consultant specializing in embedded systems and edge AI. He holds a B.A. in Mathematics from Grinnell College. His research focuses on computational and human-AI approaches to analytic number theory, particularly the zero geometry of the Riemann zeta function.

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Classification and keywords

2020 Mathematics Subject Classification: 11M26 (primary), 11M06 (secondary).

Keywords: Riemann zeta function; critical-line zeros; simple zeros; pair correlation; finite compression; inverse sampling; interval arithmetic; vertical box; zero density; artificial intelligence in mathematics.